Example 8.9 For the following ungrouped data 10, 17, 16, 21, 13, 18, 12, 10, 19, 22. Find the median.

Solution

Arrange the values in ascending order.

10, 10, 12, 13, 16, 17, 18, 19, 21, 22.

The number of values = 10

$$\text{Median} = \text{Average of } \left(\frac{10}{2}\right)^{\text{th}} \text{ and } \left(\frac{10}{2}+1\right)^{th} \text{ values}$$

$$= \text{Average of } 5^{\text{th}} \text{ and } 6^{\text{th}} \text{ values}$$

$$= \frac{16+17}{2} = \frac{33}{2} = 16.5$$

Example 8.10 The following table represents the marks obtained by a group of 12 students in a class test in Mathematics and Science.

Marks (Mathematics)	52	55	32	30	60	44	28	25	50	75	33	62
Marks (Science)	54	42	48	49	27	25	24	19	28	58	42	69

Indicate in which subject, the level of achievement is higher?

Solution

Let us arrange the marks in the two subjects in ascending order.

Marks (Mathematics)	25	28	30	32	33	44	50	52	55	60	62	75
Marks (Science)	19	24	25	27	28	42	42	48	49	54	58	69

Since the number of students is 12, the marks of the middle-most student would be the mean mark of 6th and 7th students.

$$\text{Therefore , Median mark in Mathematics} = \frac{44+50}{2} = 47$$

$$\text{Median mark in Science} = \frac{42+42}{2} = 42$$

Here the median mark in Mathematics is greater than the median mark in Science. Therefore, the level of achievement of the students is higher in Mathematics than Science.

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